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Inventor(s)	Hsu; Kuo-Chun et al.

Electrical connector assembly with improved terminals

Abstract

An electrical connector assembly includes a plug connector and a receptacle connector, the receptacle connector including: a receptacle housing defining a mating surface and plural passageways; and at least two receptacle power terminals and at least one receptacle signal terminal received in corresponding passageways, wherein the receptacle signal terminal is composed of two independent blanking pieces, each blanking piece comprises two elastic arms each having a contacting portion, the two blanking pieces are arranged crosswise and located in a same passageway, and the contacting portions intersect in an X shape along an inserting direction perpendicular to the mating surface, and wherein one of the two blanking pieces has a first extending portion extending opposite to the elastic arms, the other of the two blanking pieces has two second extending portions spaced with a gap, and the first extending portion goes across the gap and is sandwiched by the two second extending portions.

Inventors: **Hsu; Kuo-Chun (New Taipei, TW), Zhu; Jian-Kuang (Kunshan, CN), Xu; Yin-Chao (Kunshan, CN)**

Applicant: **FOXCONN (KUNSHAN) COMPUTER CONNECTOR CO., LTD.** (Kunshan, CN); **FOXCONN INTERCONNECT TECHNOLOGY LIMITED** (Grand Cayman, KY)

Family ID: **87850009**

Assignee: **FOXCONN (KUNSHAN) COMPUTER CONNECTOR CO., LTD. (Kunshan, CN); FOXCONN INTERCONNECT TECHNOLOGY LIMITED (Grand Cayman, KY)**

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Background/Summary

BACKGROUND OF THE INVENTION
Field of the Invention
(1) The present invention relates generally to an electrical connector assembly to transmit power and signal.
Description of Related Arts
(2) China Utility Model No. 211456029 discloses an electrical connector including an insulative housing and power terminals retained in the insulative housing. The housing includes a mating portion and a sleeve portion surrounding the mating portion with a ring gap. The mating portion defines a plurality of terminal passages receiving the power terminals. However, a retention force between the power terminals of the electrical connector and corresponding terminals of a mating connector is to be improved and adapted for a vibration environment.
(3) An improved electrical connector assembly is desired.
SUMMARY OF THE INVENTION
(4) An object of the present invention is to provide an electrical connector assembly with an improved terminal mating stability.
(5) To achieve the above-mentioned object, an electrical connector assembly comprises a plug connector and a receptacle connector for mating with the plug connector, the receptacle connector comprising: a receptacle housing defining a mating surface and a plurality of passageways through the mating surface; at least two receptacle power terminals received in corresponding passageways; and at least one receptacle signal terminal received in corresponding passageways, wherein the receptacle signal terminal is composed of two independent blanking pieces, each blanking piece comprises two elastic arms each having a contacting portion, the two blanking pieces are arranged crosswise and located in a same passageway, and the contacting portions intersect in an X shape along an inserting direction perpendicular to the mating surface.

Description

BRIEF DESCRIPTION OF THE DRAWING
(1) FIG. **1** is a perspective view of an electrical connector assembly according to an embodiment of the present invention, including a plug connector and a receptacle connector;
(2) FIG. **2** is another perspective view of the electrical connector assembly in FIG. **1**;
(3) FIG. **3** is an exploded view of the plug connector in FIG. **2**;
(4) FIG. **4** is an exploded view of the receptacle connector in FIG. **1**;
(5) FIG. **5** is a perspective view of a receptacle power terminal of the receptacle connector in FIG. **4**;
(6) FIG. **6** is an exploded view of the receptacle power terminal in FIG. **5**;
(7) FIG. **7** is a perspective view of the receptacle signal terminal of the receptacle connector in FIG. **4**;
(8) FIG. **8** is an exploded view of the receptacle signal terminal in FIG. **7**;
(9) FIG. **9** is a front view of the plug connector in FIG. **1**; and
(10) FIG. **10** is a front view of the receptacle connector in FIG. **1**.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT
(11) Referring to FIGS. **1-10**, an electrical connector assembly **1000** of an embodiment of this present invention includes a plug connector **100** and a receptacle connector **200** adapted to be mated with each other. The connector assembly transmits power and signal.
(12) Referring to FIGS. **2-4**, the plug connector **100** includes an insulative plug housing **1** and a plurality of conductive plug terminals **2** retained in the plug housing **1**. The plug housing **1** includes a mating cavity **11** opening forwards. The plug terminal **2** includes a columnar contacting portion **21** exposed upon the mating cavity **11**. The plug terminal **2** includes at least two plug power

terminals **24** and at least one plug signal terminal **25**.

(13) The receptacle connector **200** includes an insulative receptacle housing **3** and a plurality of conductive receptacle terminals **4** retained in the receptacle housing **3**. The receptacle housing **3** defines a mating surface **331** and a plurality of passageways **31** going through the mating surface **331**. The passageways **31** receive the receptacle terminals **4** one by one. The receptacle terminal **4** includes at least two receptacle power terminals **41** and at least one receptacle signal terminal **42**. Referring to FIGS. **7-8**, the receptacle signal terminal **42** is composed of two independent terminal pieces **46**, each terminal piece **46** includes two elastic arms **461**, and each elastic arm **461** defines a contacting portion **462** at a front end thereof. The terminal pieces **46** are made by blanking process, that is, blanking pieces. The two terminal pieces **46** are arranged crosswise and commonly retained in the corresponding passageway **31**. In an inserted direction of the receptacle connector perpendicular to the mating surface **331**, the contacting portions **462** of the two signal pieces **46** intersect in an X shape.

(14) Referring to FIG. **3**, the plug terminal **2** includes a retaining portion **22** fixed to the plug housing **1** and a soldering tail **23** extending rearwards from the retaining portion, the contacting portion **21** extends forward from the retaining portion **22**. In this embodiment, the plug terminals **2** include two plug power terminals **24** and seven plug signal terminals **25**. The plug power terminals and the plug signal terminals have a similar construct but differ in diameter. A diameter of the contacting portion of the plug power terminal **24** is larger than the diameter of the contacting portion of the plug signal terminal **25**. The contacting portion of the plug power terminal **24** includes a first contacting area **241** and a second contacting area **242** behind the first contact area **241**. The diameter of the second contact area **242** is smaller than that of the first contact area **241** to form a concave ring behind the first contact area **241**. A guiding portion **243** of which the diameter gradually decreases, extends forwards from the first contacting area **241**. The length of the guiding portion **243** is 1.8 mm. The soldering tail of the plug signal terminal **25** is cylindrical, and the soldering tail of the plug power terminal **24** is semi-cylindrical. The plug signal terminals **25** are arranged symmetrically with an imaginary line while the two power terminals **24** are located along the imaginary line.

(15) Referring to FIGS. **4-5**, the receptacle housing **3** defines the mating surface **331** and the passageways **31** going through the mating surface **331**. The receptacle terminals **4** includes contacting portions **43** retained in the corresponding passageways **31**, soldering tails **45** and connection portions **44** connected with the contacting portions **43** and the soldering tails **45**. In the inserted direction, the contacting portion **43** of the receptacle signal terminal **42** in the passageway **31** is of an X-shape. The receptacle housing **3** further includes a bottom wall **32**, a mating portion **33** extending from the bottom wall **32**, and a sleeve wall **34** surrounding the mating portion **33**. The front end surface of the mating portion **33** is defined as the mating surface **331**. The sleeve wall **34** and the mating portion **33** are spaced apart from each other with a ring cavity **35**, and the mating portion **33** defines the passageways **31** to receive the receptacle terminals **4**. The passageways **31** includes power passageways **311** to receive the receptacle power terminal **41** and signal passageways **312** to receive the receptacle signal terminals **42**.

(16) Referring to FIGS. **7-8**, each receptacle signal terminal **42** is composed of two independent terminal pieces **46**. Each terminal pieces **46** comprises two elastic arms **461**, each elastic arm **461** has a contacting portion **462** at distal free end thereof, and the two terminal pieces **46** are intersected and received in a same passageway **312**. The two elastic arms **461** of each terminal piece **46** are located on the same plane and opposite to each other. One of the two terminal pieces **46** has a first extending portion **463** extending backwards, and the other one has two second extending portions **464** extending backwards and spaced from each other with a narrow gap **465**. When the two terminal pieces **46** intersect, the first extending portion **463** goes cross the gap **465** and is sandwiched with the two second extending portions **464**.

(17) Referring to FIGS. **5-6**, the receptacle power terminal **41** is composed of three independent

terminal pieces **47**, one first pieces **471** made by a blanking process and two second piece **472** made by a forming process, the second pieces are located at opposite sides of the first piece **471**. The first piece **471** or blanking piece includes two elastic arms **473** with protruding contact portions, and the second piece **472** or forming pieces includes three elastic arms **474** with protruding contacting portions. The elastic arms **473**, **474** are arranged in a ring and located in a same passageway **31**. The second pieces **472** are symmetrical about the first piece **471**. As shown in FIGS. **2** and **4**, the receptacle connector **200** includes a fixed cover **5** retained in the power passageway **311** from back to front. The fixed cover **5** defines a slot **51** and a tuber **52** extending forwards from a middle of the fixed cover **5**. The soldering tails of first and second pieces **471**, **472** pass through the slot **51**. The fixed cover **5** covers a rear end of the power passageway **311** to prevent an extravasated glue into the power passageway **311**.
(18) The two elastic arms **473** extends horizontally along a front-to-back direction and are oppositely arranged. The elastic arms **474** extend obliquely inwards. The contacting portions **43** of the first piece **471** are located in front of the contact portions **43** of the second pieces **472**. The contacting portion **43** of the first piece **471** is in contact with the second contacting area **242** of the plug connector. The contacting portions **43** of the second pieces are in contact with the first contacting area **241**.
(19) The above-mentioned embodiments are only preferred embodiments of the present invention, and should not limit the scope of the present invention, any simple equivalent changes and modifications made according to the claims of the present invention and the contents of the description should still belong to the present invention.

Claims

1. An electrical connector assembly comprising: a plug connector; and a receptacle connector for mating with the plug connector, the receptacle connector comprising: a receptacle housing defining a mating surface and a plurality of passageways through the mating surface; and at least two receptacle power terminals and at least one receptacle signal terminal received in corresponding passageways; wherein the receptacle signal terminal is composed of two independent blanking pieces, each blanking piece comprises two elastic arms each having a contacting portion, the two blanking pieces are arranged crosswise and located in a same passageway, and the contacting portions intersect in an X shape along an inserting direction perpendicular to the mating surface; and wherein one of the two blanking pieces comprises a first extending portion extending opposite to the elastic arms, the other of the two blanking pieces comprises two second extending portions spaced with a gap, and the first extending portion goes across the gap and is sandwiched by the two second extending portions.
2. The electrical connector assembly as claimed in claim 1, wherein the receptacle connector comprises three receptacle signal terminals arranged in a row and two receptacle signal terminals at two opposite ends of the row.
3. The electrical connector assembly as claimed in claim 1, wherein the receptacle housing comprises a bottom wall, a mating portion extending from the bottom wall, and a sleeve wall surrounding and spaced from the mating portion with a ring cavity, and the passageways are disposed on the mating portion.
4. The electrical connector assembly as claimed in claim 1, wherein the receptacle power terminal is composed of a blanking piece and two forming pieces at opposite side of the blanking piece, and the blanking piece and the forming pieces are independent from each other and are ringed and received in a same passageway.
5. The electrical connector assembly as claimed in claim 4, wherein the blanking piece of the receptacle power terminal comprises two elastic arms with contacting portions, each of the two forming pieces comprises three elastic arms with contacting portions, and the elastic arms of the

receptacle power terminal are ring shaped.
6. The electrical connector assembly as claimed in claim 5, wherein the contacting portions of the blanking pieces of the receptacle power terminals are located nearer to the mating surface than the contacting portions of the forming pieces.
7. The electrical connector assembly as claimed in claim 1, wherein the plug connector comprises: a plug housing; and a plurality of plug terminals retained in the plug housing and comprising plug power terminals and plug signal terminals; wherein each of the plug terminals comprises a contacting portion of a columnar shape.
8. The electrical connector assembly as claimed in claim 7, wherein the contacting portion of the plug power terminal defines a first contact area and a second contact area behind the first contact area, and the second contact area is smaller than the first contact area in diameter so as to form a concave ring.
9. A receptacle connector comprising: a receptacle housing defining a mating surface and a plurality of passageways through the mating surface; a receptacle power terminal retained in a corresponding passageway; and a receptacle signal terminal retained in a corresponding passageway; wherein the receptacle signal terminal comprises four elastic arms each having a contacting portion, and the four contacting portions are arranged in an X shape in an inserting direction of the receptacle connector perpendicular to the mating surface; and wherein the receptacle power terminal is composed of one first piece and two second pieces located at opposite sides of the first piece, the first piece comprises two elastic arms, each of the two second pieces comprises three elastic arms, and the elastic arms of the first and the second pieces are ring shaped and received in a same terminal passageway.
10. The receptacle connector as claimed in claim 9, wherein the first piece is of a blanking piece, and the second pieces are of a forming piece.
11. The receptacle connector as claimed in claim 9, wherein the receptacle signal terminal is composed of two independent terminal pieces, each terminal piece comprises two elastic arms, and the two terminal pieces are arranged crosswise and received at a same passageway.
12. The receptacle connector as claimed in claim 11, wherein one of the two terminal pieces comprises a first extending portion extending opposite to the elastic arms, the other terminal piece has two second extending portions extending opposite to the elastic arms and spaced from each other with a gap, and the first extending portion is held by the gap.
13. The receptacle connector as claimed in claim 11, wherein the contacting portions of one of the two terminal pieces are located behind the contacting points of the other terminal piece thereof.
14. The receptacle connector as claimed in claim 11, wherein the two terminal pieces are of a blanking piece.
15. The receptacle connector as claimed in claim 11, wherein the two elastic arms of each terminal piece are located on a same plane and opposite to each other.
16. An electrical connector assembly comprising: a plug connector; and a receptacle connector for mating with the plug connector, the receptacle connector comprising: a receptacle housing defining a mating surface and a plurality of passageways through the mating surface; and at least two receptacle power terminals and at least one receptacle signal terminal received in corresponding passageways; wherein the receptacle signal terminal is composed of two independent blanking pieces, each blanking piece comprises two elastic arms each having a contacting portion, the two blanking pieces are arranged crosswise and located in a same passageway, and the contacting portions intersect in an X shape along an inserting direction perpendicular to the mating surface; and wherein the receptacle power terminal is composed of a blanking piece and two forming pieces at opposite side of the blanking piece, and the blanking piece and the forming pieces are independent from each other and are ringed and received in a same passageway.
17. The electrical connector assembly as claimed in claim 16, wherein the blanking piece of the receptacle power terminal comprises two elastic arms with contacting portions, each of the two

forming pieces comprises three elastic arms with contacting portions, and the elastic arms of the receptacle power terminal are ring shaped.

18. The electrical connector assembly as claimed in claim 17, wherein the contacting portions of the blanking pieces of the receptacle power terminals are located nearer to the mating surface than the contacting portions of the forming pieces.

19. The electrical connector assembly as claimed in claim 16, wherein the plug connector comprises: a plug housing; and a plurality of plug terminals retained in the plug housing and comprising plug power terminals and plug signal terminals; wherein each of the plug terminals comprises a contacting portion of a columnar shape.

20. The electrical connector assembly as claimed in claim 19, wherein the contacting portion of the plug power terminal defines a first contact area and a second contact area behind the first contact area, and the second contact area is smaller than the first contact area in diameter so as to form a concave ring.
